

Amit Tiwari

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EDUCATION

IIT BOMBAY

MS(R) IN ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE

2025 - ongoing | Bombay, India
CPI: XXX/10.0 (2 semesters)

IIT GUWAHATI

BTECH. IN ELECTRICAL AND
ELECTRONICS ENGINEERING (EEE)

2017 - 2021 | Assam, India
CPI: XXX/10.0

RESEARCH FOCUS

TOPIC:

Improving MLLMs ASR accuracy with
test-time based interventions

ADVISOR:

Prof. Preethi Jyothi, [CSALT Lab, IITB]

KEY COURSES

GRADUATE

Foundations of Machine Learning
Statistical Signal Analysis
Algorithms and Complexity
Optimization
Matrix Computations
Efficient AI
Theoretical Machine Learning

UNDERGRADUATE

Digital Circuits and Microprocessors
Digital Signal Processing
Fundamentals of VLSI CAD
Control System + Advanced Control
System + Controls Lab

SKILLS

PROGRAMMING

Python • C • C++ • Matlab

SCRIPTING

Bash • Python

DOCUMENTATION

L^AT_EX • MSOffice

SIMULATION ENV.

Keysight ADS • PSIM • Simulink • Ltspice

TOOLS

Qtronic Silver • AVL PUMA • Vector
CANalyser • RMDV DOORS

ACHIEVEMENTS

UNDERGRADUATE

Awarded Institute Merit Scholarship for
A.Y. 2020

PROJECTS

REPLICATING DOUBLE DESCENT

PROF ABIR DE, IIT BOMBAY

Aug 2025 – Nov 2025

- Successfully demonstrated the double descent phenomena on CIFAR-10 and MNIST dataset using Random Fourier Features. Original paper: Belkin et al., Reconciling modern machine-learning practice and the classical bias-variance trade-off, 2019
- Extended the experiment to include FCNN, CNN and ResNET

SOLAR POWERED BATTERY CHARGING SYSTEM

PROF RAVINDRANATH ADDA, IIT GUWAHATI

Sep 2020 – May 2021

- Proposed a Charging Solution for light duty electric vehicle. The project involved choosing array of electrical components, and designing a control system and power electronics design for DC-DC Converter.
- Implemented a Perturb and Oscillation algorithm in MATLAB for efficient capture of solar energy at PV panel. Also implemented safety features that preserve battery health based on the state parameters of the Li-Ion pack.

WORK EXPERIENCE

TEXAS INSTRUMENTS | VALIDATION ENGINEER, HIGH SPEED CONVERTERS

Sep 2022 – Sep 2023 | Bengaluru, India

- Design of impedance matching network for High frequency transceiver (7 GHz band)
- Characterization of AFE80XX devices.
- Development of python scripts to automate testing of RF Devices.

JAGUAR LAND ROVER INDIA | ELECTRONICS TRAINEE (VALIDATION ENGINEER), ONBOARD CHARGERS (ELECTRIC VEHICLE)

Jul 2021 – Jul 2022 | Bengaluru, India

- Developed a GUI using Python's Tkinter Library to provide inputs for MiL (Model In Loop) testing of Powertrain module.
- Automated test cases for PHEV using AVL's PUMA software (Vehicle In Loop Testing).

TRAININGS

RF AND MILLIMETER-WAVE CIRCUIT DESIGN Apr 2023

- Completed the course with 89.25% grade(certificate link below)
<https://www.coursera.org/account/accomplishments/verify/7CW2RWN3VBVV>

ECARS2X: ELECTRIC CARS: TECHNOLOGY Jul 2021

- Successfully completed the course(certificate link below) <https://courses.edx.org/certificates/9aa0a1e075234827a7b1e22c9ce92604>

INDUSTRIAL AUTOMATION, AUTOSYS INDORE Dec 2018

- Learnt to automate industrial processes using Programmable Logic Controllers.

C++, UNIVERSAL INFORMATICS Dec 2018

- Learnt C++ with Special emphasis on Object Oriented Programming.
- Cleared the test required for certification.